

1G/10G TCP IP Offload Engine

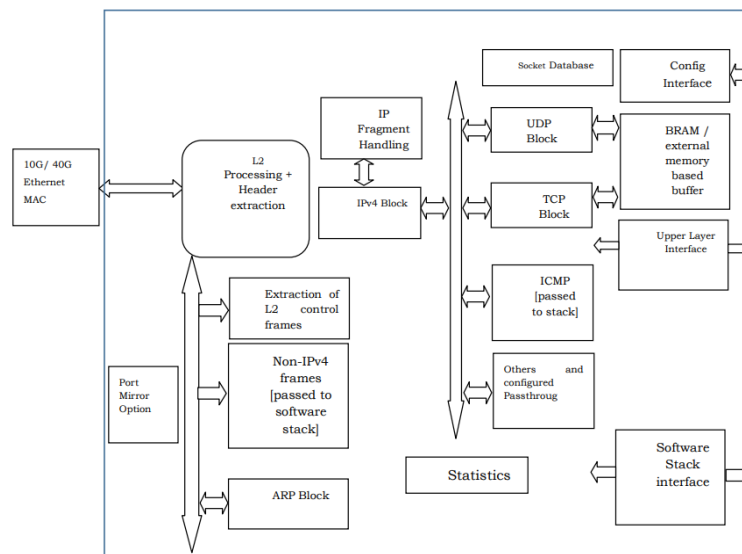
The 1G/10G TCP IP Offload Engine is a high-performance networking IP designed to accelerate TCP/IP processing in high-throughput and low-latency Ethernet systems. It offloads TCP/IP protocol handling from the host processor, enabling efficient packet processing with deterministic performance for bandwidth-intensive and real-time applications on FPGA-based platforms.

The IP core supports hardware-based TCP/IP header generation, checksum insertion and verification, fragmentation handling, and traffic classification, ensuring reliable data transfer at line rates of 1G and 10G Ethernet. Hardware-validated performance, fully synthesizable RTL, and comprehensive verification collateral make the IP engine well suited for mission-critical, industrial, and data-intensive networking applications.

Introduction

Transmission Control Protocol (TCP) offloading is a hardware acceleration technique that improves network throughput, reduces CPU load, and ensures deterministic packet processing in high-speed Ethernet systems. The 1G/10G TCP IP Offload Engine enables efficient handling of TCP/IP protocol functions in hardware, forming a critical data-path processing block for high-performance and real-time networking applications.

The IP core supports line-rate TCP processing at 1G and 10G Ethernet speeds, including header generation, checksum handling, fragmentation support, and traffic classification. Advanced packet processing features, hardware-validated performance, and seamless integration with high-speed Ethernet MACs and host interfaces enable deployment in mission-critical networking systems requiring low latency, high reliability, and deterministic operation.



Architecture

The 10G TCP & UDP IP Offload Engine interfaces with the Ethernet MAC and processes incoming frames through an L2 processing and header extraction stage. Control frames are extracted, while non-IPv4 traffic is forwarded to the software stack. IPv4 frames pass through IP fragmentation handling and protocol classification before being directed to dedicated UDP, TCP, ICMP, or pass-through processing blocks. TCP and UDP data paths interface with socket databases and BRAM or external memory buffers for high-throughput data handling. A configuration interface enables control and session management, while upper-layer and software stack interfaces handle non-offloaded traffic. Centralized statistics collection and optional port mirroring complete the architecture, enabling efficient, line-rate packet processing with minimal CPU involvement.

Why Choose Us?

 Rapid integration, faster time-to-market

 Clean, verified, documented RTL

 Direct access to experts



104, FF, Iris Tech Park, Sector 48
Sohna Road, Gurugram, 122018



+91 124 4048731



sales@qbitlabs.co.in

Features

- Complete 10G TCP and UDP Offload Engine IP Core providing hardware-accelerated networking for FPGA-based systems.
- Supports up to 32 concurrent TCP and UDP sessions, independently configurable based on available on-chip memory.
- Configurable payload buffer sizes of 8K / 16K / 64K per session for flexible performance optimization.
- Delivers line-rate 10 Gbps bi-directional throughput with deterministic latency up to ~500 ns.
- Complete protocol solution covering Layer-1 to Layer-4, with optional Layer-3 feature support.
- Compatible with Xilinx and Intel (Altera) FPGAs, supporting AXI-Stream and Avalon interfaces respectively.
- Simple 8-byte user interface with single-clock design for easy system integration.

Layer-1 / Layer-2 Capabilities

- Integration with FPGA vendor-provided PCS layer.
- Supports Ethernet flow control (Pause frames), jumbo frames, and MTU up to 9600 bytes.
- VLAN and QinQ tagging support.
- Hardware-accelerated ARP request/reply handling and ARP table management.

Layer-3 / Layer-4 Capabilities

- Supports IPv4 with optional IPv6, including option/extension header parsing.
- Hardware-based IP, TCP, and UDP checksum generation and verification.
- Supports IP fragmentation and reassembly, including multiple parallel reassembly contexts (optional).
- Advanced TCP features including segmentation/reassembly, sliding-window flow control, session creation and teardown, out-of-order packet detection, and per-session timers.

Specifications

S No	Description	Specification
1.	10G Support	• Supports 10G operation with external memory
2.	Line-Side Interface	• Line-side interface compliant with IEEE 802.3 standard
3.	System-Side Interface	• System-side data path supported through parallel bus interface
4.	IPv4 Layer Support	• Complete implementation of IPv4 protocol block
		• IP header generation based on configured parameters
		• IP header checksum compare and insert support
		• DSCP (Class of Service) marking support
		• IP fragmentation handling for large UDP PDUs



Specifications Cont...

4.	IPv4 Layer Support	Forwarding of non-configured socket frames to TCP software stack, including non-TCP/UDP frames
		Discarding of non-compliant frames
		Forwarding of IPv4 frames with extensions to upper layer (data processing only)
		Statistics support as per MIB requirements
5.UDP Layer Support		Complete UDP layer implementation
		Data checksum calculation (data + pseudo header) with configurable discard on error
		Support for UDP-Lite protocol (UDP without data checksum)
		Supports frame sizes up to 64 KB
6.	TCP Layer Support	Complete TCP layer implementation
		Socket initialization and opening in both directions based on configuration
		Support for TCP high-speed extensions as per RFC 7323 (large window size, extended timestamp)
		ACK generation, selective ACK (SACK), and retransmission support
		Automatic socket re-opening on timeout or reset events
		Supports up to 32 concurrent sockets
7.	FPGA Compatibility	Supports Xilinx and Intel (Altera) FPGA families
8.	IP Delivery	IP core delivered with complete source code (Verilog / VHDL)

Deliverables

- Basic net-list according to IP features selected
- Example Design for board validation
- Source Code (Optional)
- IP Customization & Integration support (Optional)

Evaluation

An evaluation license is available for this core. The evaluation version of the core operates in the same way as the full version for some time, dependent on clock frequency. Operation is then disabled and the data output does not change.

Get in touch with us

Delivering Advanced Solutions for
Next-Gen Networking



104, FF, Iris Tech Park, Sector 48
Sohna Road, Gurugram, 122018



+91 124 4048731



sales@qbitlabs.co.in